

A glimpse at bubbling phenomena in geometric analysis

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Introduction: A model problem Bubbling phenomena are a typical issue that arises in many *conformally invariant* problems in geometric analysis, where some energy can concentrate in a smaller and smaller region along a weakly converging sequence and disappear at the limit in what is usually called a *bubble*. They are responsible for *loss of compactness* for quantities connected to a scale invariant energy. Such phenomena are notably present in the work of P.-L. Lions [5, 6] on concentration–compactness, which also contains many references to concrete instances of these phenomena.

Let us illustrate the principle at work via the following example. We work in the *manifold-valued Sobolev space* $W^{1,n}(\mathbb{S}^n; \mathbb{S}^n)$, which consists of all $W^{1,n}$ functions u from $\mathbb{S}^n$ to $\mathbb{R}^{n+1}$ such that $u(x) \in \mathbb{S}^n$ for a.e. $x \in \mathbb{S}^n$, that is, such that $|u(x)| = 1$ for a.e. $x \in \mathbb{S}^n$. In order to produce n -harmonic mappings from $\mathbb{S}^n$ to $\mathbb{S}^n$, a natural idea is to look for solutions of the minimisation problem

$$\min_{\substack{u \in W^{1,n}(\mathbb{S}^n; \mathbb{S}^n) \\ \deg u = k}} \int_{\mathbb{S}^n} |Du|^n \quad (1)$$

for some fixed $k \in \mathbb{Z} \setminus \{0\}$, where $\deg u$ denotes the topological degree of the mappings u — intuitively, it captures how many times u warps $\mathbb{S}^n$ around itself, taking orientation into account.

This problem of looking for harmonic maps with given degree — or more generally in a given homotopy class — was already studied in the 60' by J. Eells and J. Sampson [4]. A major success of bubbling analysis — in the spirit of Theorem 3 that we present below, but involving harmonic maps rather than mere Sobolev maps and their degree — is the energy identity by J. Sacks and K. Uhlenbeck [9], which essentially allows to conclude that homotopy classes containing a minimising harmonic map form a generating set of the corresponding homotopy group. The exact bubbling structure obtained along a weakly converging sequence of harmonic maps was later on precised by T. Parker [7], using the notion of *bubble tree*. Energy identities à la Sacks and Uhlenbeck were later on a key ingredient in the proof by T. Rivière [8] of the existence of infinitely many values of the Hopf invariant for which there exists a minimising harmonic map $\mathbb{S}^3 \rightarrow \mathbb{S}^2$ having this Hopf invariant — even if the proof does not allow to compute explicitly those classes! Bubbling analysis has been used as a crucial tool in many other critical calculus of variations problems with constraint, but it would be unrealistic to try give even a partially exhaustive list of them here.

Already defining the degree for $W^{1,n}(\mathbb{S}^n; \mathbb{S}^n)$ maps — which need not be continuous in the critical dimension n — is a nontrivial task. This has been done, in a very general framework, by H. Brezis and L. Nirenberg [1], using the theory of VMO, which is the target of the limiting Sobolev embedding in critical dimension n . In our case, one can simply define the degree via the Brouwer formula

$$\deg u = \oint_{\mathbb{S}^n} \mathcal{J}u,$$

where $\oint_{\mathbb{S}^n} = \frac{1}{|\mathbb{S}^n|} \int_{\mathbb{S}^n}$ denotes the average integral, and $\mathcal{J}u = |\det Du|$ is the (absolute value of the) Jacobian determinant; see e.g. [3]. It is clear that this is well-defined for $u \in W^{1,n}(\mathbb{S}^n; \mathbb{S}^n)$ by Hölder's inequality.

A natural approach to solve (1) is via the *direct method of the calculus of variations*, denoting c the corresponding infimum and taking $(u_j)_{j \in \mathbb{N}}$ a sequence of maps in $W^{1,n}(\mathbb{S}^n; \mathbb{S}^n)$ having degree k and such that $\int_{\mathbb{S}^n} |Du_j|^n \rightarrow c$ as $j \rightarrow +\infty$. By the Banach–Alaoglu theorem, up to extraction of a (non relabelled) subsequence, one may assume that $u_j \rightharpoonup u$ in $W^{1,n}$, and by the lower semi-continuity of the Sobolev energy,

$$\int_{\mathbb{S}^n} |Du|^n \leq \liminf_{j \rightarrow +\infty} \int_{\mathbb{S}^n} |Du_j|^n = c.$$

If we would know that $\deg u = k$, we would conclude that u is an admissible competitor reaching the infimum, hence proving the existence of a solution to our problem. Unfortunately, the degree need not be preserved through weak convergence, as the following example shows.

To start with, let $v \in W^{1,n}(\mathbb{B}^n; \mathbb{S}^n)$ be a degree one map such that $v = a$ on $\partial \mathbb{B}^n$, for some fixed point $a \in \mathbb{S}^n$. Define $v_\varepsilon(x) = v(x/\varepsilon)$ on B_ε^n . By the scaling invariance of the $W^{1,n}$ energy in dimension n , we have

$$\int_{B_\varepsilon^n} |Dv_\varepsilon|^n = \int_{\mathbb{B}^n} |Dv|^n$$

— this is where we use crucially that the Sobolev exponent and the dimension coincide, making our problem conformally invariant. Now, we construct a map $u_\varepsilon \in W^{1,n}(\mathbb{S}^n; \mathbb{S}^n)$ by “patching” v_ε on a small cap of radius ε around a and sending the remaining of $\mathbb{S}^n$ to a . Hence, the maps u_ε have uniformly bounded $W^{1,n}$ energy, they have degree 1, but they weakly converge towards the map constantly equal to a , which has degree 0. What happened here is that all the topological charge concentrated in a small “bubble” around a , taking profit of the scale invariance of the problem, and vanished at the limit.

Towards a more precise bubbling analysis Being faced with a possible loss of compactness for the degree due to bubbling phenomena, it is natural to try studying more in depth those phenomena, to see if one can quantify when and how they are happening, and possibly rule them out under certain conditions. We present several statements in increasing order of complexity. We start with the following one, which rules out bubbling under a small energy assumption.

Proposition 1 (No bubbling with small energy). *For every $u \in W^{1,n}(\mathbb{S}^n; \mathbb{S}^n)$, it holds*

$$\int_{\mathbb{S}^n} |Du|^n \geq \frac{|\deg u|}{|\mathbb{S}^n| n^{n/2}}.$$

In particular, if $\int_{\mathbb{S}^n} |Du_j|^n \leq c < \frac{1}{|\mathbb{S}^n|^{n/2}}$ along a sequence weakly converging to u , then the limit u satisfies the same estimate, and hence $\deg u_j = 0 = \deg u$, ruling out any change of degree via bubbling. The proof is a direct consequence of Brouwer's formula for the degree combined with the inequality $\mathcal{J}u \leq n^{-n/2} |Du|^n$, which follows from the arithmetico-geometric inequality.

The next statement ensures that bubbling necessarily comes with some energy concentration in a small region — which is precisely the issue in our introductory example, where energy concentrates around one point.

Proposition 2 (No bubbling without concentration). *There exists $\eta > 0$ depending only on n such that, if $\rho > 0$ is sufficiently small, then $\deg u = \deg v$ for any u and $v \in W^{1,n}(\mathbb{S}^n; \mathbb{S}^n)$ satisfying*

$$\int_{B_\rho^n(a)} |Du|^n + |Dv|^n \leq \eta \quad \text{and} \quad \int_{B_\rho^n(a)} \text{dist}_{\mathbb{S}^n}(u, v)^n \leq \eta \quad \text{for every } a \in \mathbb{S}^n.$$

The proof of this proposition is by explicit construction of a homotopy connecting u and v . Namely, we define

$$u_r(a) = \int_{B_r^n(a)} u \quad \text{and} \quad v_r(a) = \int_{B_r^n(a)} v.$$

Using the smallness assumption on the gradient, we can prove that u_r and v_r remain uniformly close to $\mathbb{S}^n$ for $0 < r \leq \rho$, and can therefore be reprojected to $\mathbb{S}^n$. Using the smallness assumption on the distance between u and v , we get that u_ρ and v_ρ are uniformly close to each other, and can therefore be connected by a straight line. After reprojecting, we therefore get a homotopy by going from u to u_ρ , then v_ρ , then v .

We now present a typical precise *bubbling statement*, which is adapted from [2, Theorem 5.1].

Theorem 3. *Let $(u_j)_{j \in \mathbb{N}}$ be a sequence in $W^{1,n}(\mathbb{S}^n; \mathbb{S}^n)$ such that $\deg u_j = k$,*

$$\sup_{j \in \mathbb{N}} \int_{\mathbb{S}^n} |Du_j| < +\infty,$$

and $u_j \rightarrow u$ almost everywhere. Up to extraction, there exists a finite Radon measure μ on $\mathbb{S}^n$ and points $a_1, \dots, a_\ell \in \mathbb{S}^n$ and numbers $d_1, \dots, d_\ell \in \mathbb{Z} \setminus \{0\}$ such that

$$|Du_j|^n \rightharpoonup \mu, \quad \mu \geq |Du|^n + \sum_{i=1}^{\ell} |\mathbb{S}^n|^{n/2} d_i \delta_{a_i}, \quad \text{and} \quad \sum_{i=1}^{\ell} d_i = k - \deg u.$$

The proof of Theorem 3 combines Propositions 1 and 2 with a careful concentration–compactness analysis. Namely, by concentration–compactness, we obtain that $|Du_j|^n$ concentrates — for a given threshold of energy — around a finite number of points. Removing small balls around the concentration points, we obtain that the assumptions of Proposition 2 are satisfied on the complement of these balls, and hence no change of degree occurs over there. On these small balls, Proposition 1 allows to relate the amount of change in the degree to the minimal amount of energy that has to concentrate in each ball. Summing over these balls and using the additivity of the degree yields the conclusion. Of course, the difficulty lies in the technicalities that are omitted here.

References

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